## Marked compact isometries and fourteen further arithmetic boundary coordinates

### D.64. A compact isometry correction and actual source involutions lift a further 1930-dimensional layer [Proved]

Use the actual groups, lattices, compact projectors and maps of D.63:

$$X = \widehat{B}/(R_{50} \cap L_2), \qquad Y = X/\mathcal{K}, \qquad X_{63} = X/\mathcal{J}_{63}, \qquad \mathcal{L} = \ker(U^- \longrightarrow U_*^-). \quad \text{(D.64.1)}$$

Number the four character-pair orbits by $\ell_1 = (0,0,0,1), \ell_2 = (0,0,1,1), \ell_3 = (0,0,1,2), \ell_4 = (0,1,0,0)$. Their sizes are $8, 12, 12, 8$. The compact orthogonal decomposition is $P_a = \widetilde{E}_a \perp M_a$, with fibre ranks 32 and 16 over the unramified quadratic coefficient ring. Let

$$\phi_a : \overline{M}_a \longrightarrow \overline{P}_{*,a}, \qquad K_a = \ker\big(\mathrm{Herm}(\phi_a) : \mathrm{Herm}_{16}(\mathbb{F}_4) \longrightarrow \mathrm{Herm}_4(\mathbb{F}_4)\big). \quad \text{(D.64.2)}$$

Bars on lattices denote reduction modulo two; on a field coefficient they denote its square. Here $\mathrm{Herm}(\phi)(B) = \phi B \bar{\phi}^t$. The actual semilinear pair stabilizer is $H_a \le Q$. Put $C_a = \mathrm{Ind}_{H_a}^{Q} K_a$, with the inflated action of the full arithmetic group $G$. There are actual onto maps

$$q_a : \mathcal{J}_{63} \longrightarrow C_a \quad \text{(D.64.3)}$$

given by compactification and the $M_a$-self projection. Their joint map is onto. Their dimensions and full group norms are

| $a$ | $\dim C_a$ | $\dim (C_a)_G$ | $\mathrm{rank}\, \nu_a$ | $\dim \ker \nu_a$ on coinvariants |
|---|---|---|---|---|
| 1 | 1920 | 4 | 4 | 0 |
| 2 | 2880 | 10 | 0 | 10 |
| 3 | 2880 | 10 | 0 | 10 |
| 4 | 1920 | 4 | 2 | 2 |

(D.64.4)

The norm is $\nu_a(v) = \sum_{g \in G} gv$, viewed either on $C_a$ or on its coinvariants as indicated.

For both actual D.57 arithmetic branches, the two old projective modules $M_1, M_4$ have explicit equivariant modulo-two isometries $X_a$ satisfying

$$\phi_a X_a = T_{*,a} \phi_a, \qquad X_a^t H_a^{\mathrm{int}} \bar{X}_a = H_a^{\mathrm{int}}, \qquad \det X_a \neq 0. \quad \text{(D.64.5)}$$

The matrix $H_a^{\mathrm{int}}$ is the reduction of the actual compact intersection form, computed from marked integral cycle lifts. The superscript distinguishes this matrix from the stabilizer group $H_a$. The complete affine parameter spaces and all their quadratic isometry equations are retained in the certificate. Their dimensions are eight and six, respectively. The two eight-parameter solutions use masks $\mathtt{c3}, 0$ for branches one and three; both six-parameter solutions use mask zero. The fixed-seed search trial counts are $62, 1, 1, 1$. All sixteen D.57 framing choices are retained.

For $a = 2, 3$, let $I_a \subset (C_a)_G$ be the four-dimensional image of the actual source-involution map

$$v \longmapsto [e_\Delta(v \wedge sv)]_G, \qquad v \in \ker \phi_a \text{ and its induced coset copies.} \quad \text{(D.64.6)}$$

There are explicit retractions $p_a : (C_a)_G \to I_a$ that intertwine every equivariant compact homology action with either of the two actual D.57 small actions. Thus their kernels are invariant under the actual retained homology action. Define the onto equivariant map

$$\begin{aligned} \Phi_{64} : \mathcal{J}_{63} &\longrightarrow \mathcal{E}_{64} = C_1 \oplus I_2 \oplus I_3 \oplus \nu_4(C_4), \\ z &\longmapsto (q_1 z,\ p_2[q_2 z],\ p_3[q_3 z],\ \nu_4 q_4 z), \\ \mathcal{J}_{64} &= \ker \Phi_{64}, \qquad X_{64} = X/\mathcal{J}_{64}. \end{aligned} \quad \text{(D.64.7)}$$

The last three summands of $\mathcal{E}_{64}$ have trivial $G$-action. The first has the induced action specified in (D.64.2)–(D.64.3). In particular

$$\dim \mathcal{E}_{64} = 1930, \qquad \dim(\mathcal{E}_{64})_G = 14, \qquad \ker(X_{64} \longrightarrow X_{63}) = \mathcal{E}_{64}. \tag{D.64.8}$$

There is an actual continuous peripheral automorphism $\alpha_{64}$ of $X_{64}$, of multiplier one, fixing the original cusp $t$ and inducing identity on $G$. It retains the complete D.57 action and its literal compatible conjugators, the full $\mathcal{W}_h$ action, and every established metabelian and CM comparison through that map. Its projection to $X_{63}$ is a new representative, with the following exact comparison on $Y$. Write $T_{63}$ for D.63's integral homology action. There are integral equivariant maps $A_a : M_a \to \mathcal{L}$, for $a = 1, 4$, and continuous projective coefficient derivations $\eta_a : Y \to M_a$, inflated from the actual full homology source $\mathcal{J}_\infty$. Their restrictions to $U = \ker(\mathcal{J}_\infty \to G)$ are $m_a$, and $\eta_a(t) = 0$. They satisfy

$$\begin{aligned} T_{64} &= T_{63} + A_1 m_1 + A_4 m_4, \\ \beta_Y(g) &= \big(A_1\eta_1(g) + A_4\eta_4(g)\big)\alpha_{63,Y}(g), \\ \alpha_{64,Y} &= \mathrm{Int}(b_{t,Y})\beta_Y, \qquad b_t = \prod_j \prod_{k=0}^{35} t^k u_j t^{-k}, \quad u_j \in S_{57}. \end{aligned} \tag{D.64.9}$$

The reductions of the compact restrictions of $T_{64}$ are the isometries $X_1, X_4$. Integral invertibility and equivariance are proved; no equality of integral intersection forms is asserted. The vectors used in $b_t$ are the earlier $\widetilde{E}$-self corrections. The eight new corrections in (D.64.6) centralize the source involution on $Y$, so they cause no additional change in (D.64.9).

The complete remaining source kernels are

$$\begin{aligned} \ker(X \to X_{64}) = \mathcal{J}_{64}, \qquad \ker(X_{64} \to Y) &= \mathcal{K}/\mathcal{J}_{64}, \\ \dim \mathcal{J}_{64} &= \dim \mathcal{J}_{63} - 1930, \\ \dim(\mathcal{J}_{64})_{\text{specified small characters}} &= 282230, \\ S_{64} &= \{b \in R_{50} : \mathrm{im}_{\mathcal{T}_2}(b) \in \mathcal{J}_{64}\}, \\ S_{64}/(R_{50} \cap L_2) &= \mathcal{J}_{64}. \end{aligned} \tag{D.64.10}$$

This is a proper, nonzero remaining source kernel. The tested involution and full-cusp maps on the entire compact kernel have combined rank eight in its 22-dimensional norm kernel. Fourteen coordinates of that particular coinvariant test remain. This count is not used as a coinvariant-dimension formula for $\mathcal{J}_{64}$. The remaining arithmetic boundary and the full profinite realization are unresolved. The manuscript passages are Part D(e), section 51.16, folio 734 verso, scan 174, and section 51.15, folios 725–727, scans 156–160.

*Proof. The actual compact maps and the residual quotient.* Use the four compact decompositions of D.63, including their proved graph maps to the earlier markings. The affine cusp lattice is $\mathcal{R}$, and $\pi : U \to P = U/\mathcal{R}$ is compactification. The small maps $\theta_U, \theta_P$ fit into the two actual exact sequences of cusp and compact homology. The source cusp has width 36 and its small image has width 12. Thus an oriented boundary loop maps to three times its small boundary loop. Every small cusp has a preimage, and three is a unit in $\mathbb{Z}_2$. It follows, with the single sum relation retained, that

$$\theta_{\mathcal{R}} : \mathcal{R} \twoheadrightarrow \mathcal{R}_*, \qquad \mathcal{L} \twoheadrightarrow \ker \theta_P. \tag{D.64.11}$$

The integral map $\theta_U$ is onto by D.58.2; the second assertion is the resulting kernel map in the exact-sequence diagram. All these kernels are saturated. Their reductions are therefore the kernels of the reduced maps. The same statements hold on the selected odd-character parts.

On an unordered nonzero character pair, odd averaging identifies the self exterior module with Hermitian matrices over $\mathbb{F}_4$. In the basis $u_i, \omega u_i$ the diagonal coordinate is $u_i \wedge \omega u_i$; the two off-diagonal coordinates are

$$\begin{aligned} H_{ij} = 1: &\quad u_i \wedge \omega u_j + \omega u_i \wedge u_j, \\ H_{ij} = \omega: &\quad u_i \wedge u_j + \omega u_i \wedge u_j + \omega u_i \wedge \omega u_j. \end{aligned} \tag{D.64.12}$$

The actual induced exterior map is consequently $B \mapsto \phi_a B \bar{\phi}_a^t$. Each $\phi_a$ has rank four, so this Hermitian map has rank 16 and kernel dimension 240 over $\mathbb{F}_2$. Equivariant induction multiplies this dimension by the corresponding orbit size.

Compactification of a vector of $\mathcal{J}_{63}$ has zero small image. Its remaining compact self part is in the four $M_a$-blocks, hence gives (D.64.3). To prove onto, start with any vector of $K_a$ and lift it into the $M_a$-self block of the full central module. Its small affine image has zero compact image, hence is in $\mathcal{R}_* \wedge \overline{U}_*$. Both maps in (D.64.11) and $\theta_U$ are onto, so subtract a lift in $\overline{\mathcal{R}} \wedge \overline{U}$. Every selected compact projector kills this lift. Odd averaging and the character projector retain the chosen orbit. The result lies in $\mathcal{J}_{63}$, with the prescribed compact self coordinate. The same construction independently on the four orbits proves joint surjectivity.

The program `d_residual_compact_norm.py` computes each complete semilinear action on the 256-dimensional Hermitian module. It restricts the action to the actual 240-dimensional kernel, forms all relations $(g-1)K_a$, and sums every stabilizer action. It checks that this sum kills all relations and has invariant image. Coinvariants and invariants of the induced module are identified with those of this fibre by the literal chosen coset representatives. Under these identifications the group norm is the stabilizer norm. The translation subgroup and the remaining odd kernel have odd order and act trivially on the averaged self module, so their norm factors are one modulo two. These exact maps give (D.64.4), including all four small comparison maps.

*The integral ribbon pairing and its arithmetic marking.* Let $\mathcal{O}/4 = \mathbb{Z}_2[\omega]/4$, with $\omega^2+\omega+1 = 0$. The graph has the actual edges indexed by $Q$, with the two vertex sets $Q/\langle s\rangle$ and $Q/\langle r\rangle$. The cyclic order at the latter is $g, gr, gr^2$. For the character $\ell$, put $k_g = \ell(Q_g v_r)$ in $\mathbb{F}_3$. The twisted incidence matrix uses coefficient one at the two-valent vertex and the consecutive coefficients $1, \omega^{-k_g}, \ldots$ at a three-valent vertex. These are the same edge and vertex maps as D.60, now calculated modulo four.

Let $U_0$ be the retained modulo-two cycle matrix, and $\widetilde{U}_0$ its coefficientwise lift to $\mathcal{O}/4$. The incidence map is onto on this nontrivial character part. Solve

$$B_2 D = \tfrac{1}{2} B_4 \widetilde{U}_0, \qquad U_4 = \widetilde{U}_0 + 2\widetilde{D}. \tag{D.64.13}$$

The certificate verifies $B_4 U_4 = 0$ and $U_4 \bmod 2 = U_0$. These cycles lift to actual integral cycles: the incidence cokernel vanishes, and the same correction solves the successive integral equations. The computation never divides a non-even coefficient by two.

For two integral cycles $u, v$, the signed half-ribbon formula, in these vertex markings, is

$$h(u,v) = \frac{1}{2} \sum_{g \in Q/\langle r\rangle} \left( u_g \omega^{k_g} \bar{v}_{gr} - u_{gr} \omega^{-k_g} \bar{v}_g \right). \tag{D.64.14}$$

Indeed displace one cycle to the two sides of the oriented ribbon. At a three-valent vertex the difference of the two crossing counts is the displayed determinant; the sum-zero condition eliminates the third edge. A two-valent vertex contributes zero. Averaging the two signed counts gives (D.64.14). The crossing number is integral, so the displayed numerator is even. This proves that the formula computes the actual intersection pairing. Its reduction depends only on the cycle coordinates modulo four. The Fourier sum over the 81 translations multiplies this expression by 81, which is one modulo four.

The odd quotient from $G$ to $G^\dagger$ has degree $d = 3^{93}$. The actual pushforward and transfer obey $\theta \operatorname{tr} = d$, and

$$\langle d^{-1} \operatorname{tr} u, d^{-1} \operatorname{tr} v \rangle_G = d^{-1} \langle u, v \rangle_{G^\dagger}. \tag{D.64.15}$$

This is the normalized transfer identification used on the selected character lattice. Its factor is one modulo two. Thus the finite marked calculation is also the reduction of the actual arithmetic intersection form on these summands, with the specified map.

The program `d_residual_compact_pairing.py` implements (D.64.13)–(D.64.14). Its affine pairing has rank 48 on each 96-dimensional fibre and kills exactly the 48-dimensional cusp subspace. The small affine pairing has rank four on its eight-dimensional fibre. The actual inclusion of that fibre in D.57's complete 649-dimensional marked cycle space satisfies

$$E^t \Omega_{649} \bar{E} = h_*. \tag{D.64.16}$$

Every entry is checked. On each $M_a$ the induced form has rank 16. Its restriction to $\ker \phi_a$ has rank eight and radical rank four. The adjoint $\overline{(H_a^{\mathrm{int}})^{-1} \phi_a^t H_{*,a}^{\mathrm{int}}}$ has rank four and is killed by $\phi_a$. All semilinear stabilizer maps preserve the actual form, with coefficient conjugation on character-exchanging elements.

Finally, applying each complete D.57 homology matrix to $E$ gives the actual small affine action and its compact quotient $T_{*,a}$. The extension certificate verifies that all eight framings in each branch have identical complete homology matrices. Both branches are therefore compared through their actual marked maps in (D.64.16), without choosing an unrelated symplectic model.

*The complete isometry equations.* For $a = 1, 4$, the unknown matrix $X$ has 512 binary entries. If $A_g$ is a stabilizer matrix and $\epsilon_g$ records character exchange, its exact equations are

$$A_g X^{[\epsilon_g]} = X A_g, \qquad \phi_a X = T_{*,a} \phi_a, \tag{D.64.17}$$

where $X^{[0]} = X$ and $X^{[1]} = \bar{X}$. The linear ranks are 504 and 506. The certificate retains a particular solution $X_0$ and every homogeneous basis matrix $D_i$, so all solutions are exactly $X_0 + \sum_i e_i D_i$, $e_i \in \mathbb{F}_2$. Substitution into the isometry equation gives the complete vector polynomial

$$c + \sum_i l_i e_i + \sum_{i<j} q_{ij} e_i e_j = 0 \quad \text{in } \operatorname{Herm}_{16}(\mathbb{F}_4). \tag{D.64.18}$$

Every coefficient, including every zero coefficient, is retained. Their spans have dimensions four and two. The masks and trial counts in the statement give literal zeros. Direct multiplication then checks all 72 stabilizer equations, rank 16, the marked small square, and the entire isometry matrix. No sampling assertion is used to certify these four solutions.

*The integral correction and its actual source maps.* The old $M_1, M_4$ lattices are projective by D.60 and remain the same lattices in D.63's orthogonal graph decomposition. Write $D_a$ for the existing compact restriction of $T_{63}$ to $M_a$. Both $\bar{D}_a$ and the chosen $X_a$ satisfy the same small square, so

$$X_a - \bar{D}_a : \overline{M}_a \longrightarrow \ker \bar{\theta}_P \tag{D.64.19}$$

is an actual equivariant map. Projectivity lifts it to an integral map through (D.64.11). Reduction is onto here: both the kernel and its quotient are saturated free lattices, and applying $\operatorname{Hom}_G(M_a, -)$ preserves the surjections. Apply the orbit projector and then $1 - e$, where $e$ is the total $\widetilde{E}$-projector. This gives $A_a : M_a \to \mathcal{L}$ with $eA_a = 0$, the required reduction, and compact image in the same $M_a$. These are exact integral maps; successive choices of lifts modulo $2^n$ are compatible by the same surjections and define their continuous inverse limits.

The projective coefficient module $M_a$ has zero positive cohomology over $G$. Inflation–restriction therefore lifts the actual affine projection $m_a : U \to M_a$ to a derivation of the full

homology source. It kills the cusp vector $[t^{36}]$. Projectivity for the actual cyclic subgroup $\langle t\rangle$ gives the integral equality $\ker N_{36}(t) = (1-t)M_a$. Subtracting its corresponding inner derivation makes the value at $t$ zero. The same exact cyclic sequences give

$$\eta_a(s) = (1-s)d_{a,s}, \qquad \eta_a(r) = (1-r)d_{a,r}, \qquad \eta_a(t) = 0. \tag{D.64.20}$$

The homology kernel is killed, so the derivation factors through $Y$, as in D.63.22. Composition with $A_a$ takes values in its actual abelian subgroup $\mathcal{L}$. Because $\alpha_{63,Y}$ induces identity on $G$, the second line of (D.64.9) is a homomorphism. The two conjugators change by the actual vectors $A_a d_{a,s}, A_a d_{a,r}$. They have zero images in $X_*$ and $\mathcal{W}_h$, and each has a genuine lift in $S_{57}$ by D.58.2.

The homology matrix is the first line of (D.64.9). It is identity on $\widetilde{E}$, preserves all cusp vectors, has the invertible reductions $X_1, X_4$ on the two changed compact blocks, and has its old invertible blocks on the others. It is therefore an integral automorphism. The quotient and kernel maps in D.58's actual fibre product are automorphisms, so $\beta_Y$ is an automorphism of $Y$. This proves the passage from the finite isometries to actual integral peripheral source maps.

*A stable quotient for the eight source-involution coordinates.* For $a = 2, 3$, repeat (D.64.17) with the actual small action. The full affine space has dimension 16 in each branch; its linear rank is 496. On the ten-dimensional space $(C_a)_G$, the exterior action is the quadratic matrix polynomial

$$F_0 + \sum_{i=1}^{16} e_i F_i + \sum_{1\le i<j\le 16} e_i e_j F_{ij}. \tag{D.64.21}$$

All 137 coefficient matrices in each branch, and all 274 in the two branches together, are retained. Naturality of (D.64.6) shows that its image $I_a$ is invariant under every one of these matrices; the program checks each of these restrictions directly as well.

In the retained ten-coordinate bases, and the ordered four-coordinate bases of $I_a$, the columns of the retractions are the following binary masks, written in hexadecimal:

$$\begin{aligned} p_2 &= (\mathtt{a}, \mathtt{5}, 0, \mathtt{2}, \mathtt{7}, \mathtt{6}, \mathtt{9}, \mathtt{2}, \mathtt{5}, 0),\\ p_3 &= (\mathtt{2}, \mathtt{2}, \mathtt{3}, \mathtt{9}, \mathtt{f}, \mathtt{5}, \mathtt{d}, \mathtt{4}, 0, 0). \end{aligned} \tag{D.64.22}$$

The inclusion columns are respectively $(\mathtt{136}, \mathtt{75}, \mathtt{2c}, \mathtt{1e0})$ and $(\mathtt{254}, \mathtt{353}, \mathtt{2bc}, \mathtt{2ab})$. Here a mask lists binary coordinates from its least significant bit upward. These inclusions are the actual independent source-s correction columns, not arbitrarily chosen subspaces.

The simultaneous equations

$$p_a i_a = 1, \qquad p_a F = F|_{I_a}\, p_a \quad \text{for every coefficient in (D.64.21)} \tag{D.64.23}$$

are linear in the 40 entries of $p_a$. They have rank 34 and are satisfied by (D.64.22). It follows by polynomial substitution that the same retraction intertwines every compact homology action in either complete affine family. In particular it intertwines the actual $T_{63}$, which remains unchanged on these two compact blocks. Its kernel is consequently invariant, with an invertible quotient action. This proves the needed descent through the eight-dimensional quotient without an assumption about the unrecorded full homology matrix.

*All actual source-involution and full-cusp terms.* The program `d_residual_compact_cusp.py` uses $\ker \phi_a$, of fibre dimension 12 over $\mathbb{F}_4$, as the source of the correction maps. It induces this module with the actual coset representatives and actual affine generators

$$s = (0, S), \qquad r = (v_r, R), \qquad t = (Q_S v_r, SR), \qquad Q_S v_r = (1, 1, 0, 1). \tag{D.64.24}$$

It checks $s^2 = r^3 = 1$ and $t = sr$ on every column. It uses the full sum $N_{36}(t) = \sum_{j=0}^{35} t^j$, retaining every term, even though this coefficient action has period 12. In particular $t^{18}$ has

Q-state 314 and translation $(0, 2, 1, 0)$; the pure Q-state 314 is not substituted for this actual cusp power.

Formula (D.64.12) gives the averaged wedge of two real vectors as $v\bar{w}^t + w\bar{v}^t$. All 496 real basis wedge pairs are checked against the literal sum over scalar multiplications by $1, \omega, \omega^2$. After taking the actual kernel coinvariants, the computed correction maps are

$$v \longmapsto [e_\Delta(v \wedge sv)]_G, \qquad v \longmapsto [e_\Delta(v \wedge N_{36}(t)v)]_G. \tag{D.64.25}$$

All 238656 polar equations vanish. The first map factors through the actual $1 + s$ norm; the second factors through the actual full $N_{36}(t)$ norm. Their ranks are

| $a$ | $\dim\ker\nu_a$ | rank($s$ correction) | rank($t$ correction) | uncovered tested coordinates |
|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 |
| 2 | 10 | 4 | 0 | 6 |
| 3 | 10 | 4 | 0 | 6 |
| 4 | 2 | 0 | 0 | 2 |

(D.64.26)

On the two retracted quotients, actual source indices $(144, 145, 147, 149)$ and $(1, 2, 3, 4)$ map to the four standard basis vectors, respectively. All indices are zero based in the retained induced source bases.

To realize these columns in the actual group, lift a chosen compact vector through (D.64.11) into $\mathcal{L}$, and then to $u \in S_{57}$. Let $\widetilde{s}$ be the actual image of the source involution in a preliminary peripheral lift. With the convention $[a, b] = aba^{-1}b^{-1}$, put

$$b_s = u\widetilde{s}u\widetilde{s}^{-1}, \qquad b_s\widetilde{s}b_s^{-1}\widetilde{s}^{-1} = [u, \widetilde{s}u\widetilde{s}^{-1}]. \tag{D.64.27}$$

The image of $b_s$ in $Y$ is the s-fixed vector $v + sv$. Changing this source conjugator therefore preserves the $Y$-action exactly. Its class-two error has precisely the first image in (D.64.25). Both comparison conjugators in $X_*$ and $\mathcal{W}_h$ remain literally unchanged. Compact lifts in these two $M$-blocks have no $\widetilde{E}$-component, so their errors have zero projections to all earlier selected compact blocks and to the other two compact self blocks.

For completeness the full cusp word is $b_t = U_0U_1 \cdots U_{35}$, where $U_j = t^jut^{-j}$, and $c = t^{36}$. Exact class-two collection gives

$$b_tb_t^{-1}t^{-1} = [u, U_1 \cdots U_{35}][c, u]^{-1}. \tag{D.64.28}$$

The wrap term is $U_{36} = cuc^{-1}$. Compactification kills the cusp factor in $[c, u]$, and the remaining term is the second map in (D.64.25). The certificate keeps this entire 36-term map; its value on the compact kernels is zero. Earlier $\widetilde{E}$ corrections still use these full words. They give exactly the inner change in (D.64.9), and their projection to every $M$-self quotient here is zero.

*Boundary zero, quotient descent and the remaining kernel.* After the homology correction, the compact action remains block diagonal for $\widetilde{E}_a \perp M_a$, with identity on the first part. All earlier fixed-row norm arguments remain valid. The D.62 self corrections are re-applied in the actual $\widetilde{E}$-lifts; they leave the new compact quotient coordinates unchanged. The earlier moving components are corrected by their actual augmentation equations, which also have zero image in the new compact quotient. This produces a preliminary source lift whose cusp defect $z$ belongs to $\mathcal{J}_{63}$.

D.55's unquotiented orientation identity applies to these literal conjugators. On an old compact self block its orientation bivector is represented by $\overline{(H_a^{\mathrm{int}})^{-1}}$. Equation (D.64.5) implies

$$X_a\overline{(H_a^{\mathrm{int}})^{-1}}\bar{X}_a^t = \overline{(H_a^{\mathrm{int}})^{-1}}, \qquad \nu_a(q_az) = 0 \quad (a = 1, 4). \tag{D.64.29}$$

The last program checks this complete bivector identity, all 256 norm-intertwining columns, and the invertible induced kernel and norm-image actions in both branches. By (D.64.4), the coinvariant class of $q_1 z$ is zero. The $\nu_4(C_4)$-coordinate itself is zero.

Use the eight actual involution words in (D.64.27) to cancel the two coordinates $p_2[q_2 z]$, $p_3[q_3 z]$. They do not change (D.64.29). Before odd averaging, these eight words can also introduce a new component in $\mathcal{K}_{\mathrm{mov}} = (1 - e_\Delta)\mathcal{K}$. Remove that component at this point, after the eight words. The group $\Delta$ is normal of odd order, so $(\mathcal{K}_{\mathrm{mov}})_\Delta = 0$ and hence $(\mathcal{K}_{\mathrm{mov}})_G = 0$. The exact two-conjugator map on this actual submodule is therefore onto:

$$(1 - s)\mathcal{K}_{\mathrm{mov}} + s(1 - r)\mathcal{K}_{\mathrm{mov}} = \mathcal{K}_{\mathrm{mov}}.$$

This is the same augmentation argument as D.58.17, now with the specified $\Delta$-moving summand in D.63.10. The full kernel proof in D.58 identifies $\mathcal{K}$ with the actual image of $R_{50} \cap L_{*,2}$ in $\mathcal{T}_2$. Thus the two chosen coefficients have source lifts in

$$R_{50} \cap L_{*,2} \ \subset \ S_{57} = \mathfrak{R}_h \cap M_* \cap L_{*,2}.$$

They have zero images in $Y$, in both literal $X_*$ conjugators, and in $\mathcal{W}_h$. All previously selected invariant projectors kill $\mathcal{K}_{\mathrm{mov}}$. The maps defining $\Phi_{64}$ also kill it, when extended by zero from the invariant summand: each begins with the compact self projection after $e_\Delta$. Consequently this post-word correction removes precisely the reintroduced moving component and preserves all the just-cancelled compact coordinates. The cusp error is again in $\mathcal{J}_{63}$.

The only remaining defect in $\mathcal{E}_{64}$ is the augmentation part of $C_1$. Since the actual generators $s, r$ generate $G$, the exact map

$$(b, d) \longmapsto (1 - s)b + s(1 - r)d \tag{D.64.30}$$

has image the augmentation submodule. Lift its chosen coefficients through the onto map $\Phi_{64}$. They are actual elements of $\mathcal{J}_{63}$, so these final central changes preserve $Y$ and every previous comparison. The cusp defect now lies in $\mathcal{J}_{64}$.

The induced exterior action preserves $\mathcal{J}_{63}$. It preserves the kernel of $q_1$ and induces an automorphism on $C_1$, because the actual compact action preserves $M_1$ and its small square. It commutes with the norm, hence preserves the kernel of $\nu_4 q_4$ and induces an automorphism on its image. The universal identities (D.64.23) give the two remaining kernel preservations. Consequently it preserves their exact intersection $\mathcal{J}_{64}$. The class-two laws and the invertible Frattini map give the automorphism on the finite class-two factor. Its common quotient with the full homology and $\mathcal{W}_h$ sources stays equivariant. D.58's marked fibre product is exactly $X/\mathcal{J}_{64}$, proving the asserted actual automorphism.

Joint surjectivity in (D.64.3), the retractions, and the two-dimensional norm image prove (D.64.8) and the dimension statements in (D.64.10). No direct-sum splitting of $\mathcal{J}_{63}$ is required: this theorem uses the actual quotient map $\Phi_{64}$. For example $C_4$ has a 1918-dimensional kernel of its norm. A nonzero vector there has a lift in $\mathcal{J}_{63}$ with zero other orbit coordinates; that lift belongs to $\mathcal{J}_{64}$ and is nonzero. This proves the remaining kernel assertion with its literal source map.

The six programs retain the integral cycle lifts, all finite coefficient matrices, complete polynomial equations, actual source actions, kernel bases, retractions and extension comparisons. Their six complete author runs and separate copied-input replays are recorded alongside the actual stdout. The fourteen uncovered coordinates in (D.64.26) are an exact limit of those two tested maps. No conclusion about the rest of the nonzero kernel in (D.64.10), or about a full $\widehat{B}$-automorphism lifting this action, is drawn from the finite calculations. □